R16

Code No: 132AA

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech I Year II Semester Examinations, May - 2019
ENGINEERING PHYSICS – II
(Common to EEE, ECE, CSE, EIE, IT, ETM)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART- A

(25 Marks)

1.a) Explain dual nature of light. [2]
b) Define de-Broglie's hypothesis. [3]
c) Define Fermi energy level at 0K. [2]
d) Draw V-I Characteristic curve of PN junction diode. [3]
e) Define Polarization vector. [2]
f) Explain pyro electricity. [3]
g) Define magnetic susceptibility. [2]
h) Distinguish between soft and hard magnetic materials. [3]
i) What is nano size? Explain briefly. [2]
j) Write any two applications of nanomaterials. [3]

PART-B

(50 Marks)

2.a) With neat diagram explain Davisson and Germer's experiment.
b) Describe formation of energy bands in solids. [5+5]

OR

3.a) Derive Schrödinger's time independent wave equation.
b) Explain physical significance of wave function (ψ). [5+5]

4.a) Distinguish between direct and indirect band gap semiconductors.
b) Determine the concentration of the charge carriers in conduction band of intrinsic semiconductors. [4+6]

OR

5.a) Explain formation of PN junction diode.
b) Describe construction and working principle of Solar Cell. [4+6]

6.a) Derive an expression for Clausius-Mosotti relation.
b) Describe Ferro electricity of a dielectric material. [5+5]

OR

7.a) Derive an expression for electronic polarizability.
b) Discuss Piezo electricity of a dielectric material. [5+5]

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8.a) Discuss classification of magnetic materials.
b) Explain hysteresis curve based on domain theory of ferro magnetism. [5+5]

OR

9.a) What is Meissner's effect? Write any two applications of superconductors.
b) Derive an expression for Bohr magneton. [5+5]

10.a) Explain how surface to volume ratio varies in nano materials.
b) Describe preparation of nanomaterials by using Sol-Gel method. [4+6]

OR

11.a) Explain Quantum confinement.
b) Explain how the nanomaterials are characterized by using SEM and TEM. [4+6]

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